Analytical Solutions for Benchmarking

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Simple generators with source

Scalar function

For the equation

$$\frac{df}{dt} = -af + b\cos(\omega t)$$

$$f(t) = e^{-at} f(0) + b \int_0^t e^{-a(t-\tau)} \cos(\omega\tau)\, d\tau$$

$$= e^{-at} f(0) + \frac{b}{2} e^{-at} \int_0^t e^{a\tau} \left(e^{i\omega\tau} + e^{-i\omega\tau}\right) d\tau$$

$$= e^{-at} f(0) + \frac{b}{2} \left(\frac{e^{i\omega t} - e^{-at}}{a + i\omega} + \frac{e^{-i\omega t} - e^{-at}}{a - i\omega} \right)$$

or in the limit of a constant source $\omega \to 0$, we have

$$f(t) = e^{-at} f(0) + \frac{b}{a} \left(1 - e^{-at}\right)$$

Vector function

$$\frac{d\mathbf{f}}{dt} = i\lambda M \mathbf{f}(t) + \mathbf{v}(t)$$

with

$$M = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

and

$$\mathbf{v}(t) = \begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix}$$

$$\mathbf{f}(t) = e^{i\lambda M t}\mathbf{f}(0) + \int_0^t e^{i\lambda M(t-\tau)}\mathbf{v}(\tau)\,d\tau$$

$$e^{-i\lambda M t} = \cos(\lambda t)\,I + i\sin(\lambda t)\,M$$

$$\int_0^t e^{i\lambda M(t-\tau)}\mathbf{v}(\tau)\,d\tau = \int_0^t \left(\cos(\lambda(t-\tau))\begin{bmatrix} v_1(\tau) \\ v_2(\tau) \end{bmatrix} - i\sin(\lambda(t-\tau))\begin{bmatrix} v_2(\tau) \\ v_1(\tau) \end{bmatrix}\right) d\tau$$

taking $v_1(t) = v_0 e^{-\alpha t}$, $v_2(t) = 0$, we have

$$= v_0 \int_0^t \left(\begin{bmatrix} \cos(\lambda(t-\tau))\,e^{-\alpha\tau} \\ -i\sin(\lambda(t-\tau))\,e^{-\alpha\tau} \end{bmatrix}\right) d\tau$$

$$= \frac{v_0}{2}\begin{bmatrix} \frac{1}{i\lambda-\alpha}\left(e^{-\alpha t} - e^{i\lambda t}\right) + \frac{1}{-i\lambda-\alpha}\left(e^{-\alpha t} - e^{-i\lambda t}\right) \\ -\frac{1}{i\lambda-\alpha}\left(e^{-\alpha t} - e^{i\lambda t}\right) + \frac{1}{-i\lambda-\alpha}\left(e^{-\alpha t} - e^{-i\lambda t}\right) \end{bmatrix}$$

or in the limit of a constant source $\alpha \to 0$, we have

$$\int_0^t e^{i\lambda M(t-\tau)}\mathbf{v}(\tau)\,d\tau = \frac{v_0}{\lambda}\begin{bmatrix} 1 - \sin(\lambda(t-\tau)) \\ i\left(1 - \cos(\lambda(t-\tau))\right) \end{bmatrix}$$

Propagating wave-packet

The initial state for a wavepacket traveling to the right

$$p(x, t=0) = f(x)$$

and the derivative is

$$\frac{dp}{dt}|_{t=0} = f'(x)\,\frac{d(x-ct)}{dt} = -cf'(x)$$

Therefore the initial state is

$$v(x, t=0) = \begin{bmatrix} f(x) \\ -cf'(x) \end{bmatrix}$$

Analytical solution for the adjoint wave equation

Assume that the speed field is constant:

$$m(x) = c^2(x) = c^2$$

The solution of the wave equation for an initial condition is just

$$v(x,t) = \begin{bmatrix} f(x-ct) \\ -cf'(x-ct) \end{bmatrix}$$

meaning that the pressure measured by the detector at position $x = x_d$ is just

$$P(x_d, t) = f(x_d - ct)$$

Lets say we guess an initial guess $m_g(x) = c_g$, the predict measurements are therefore

$$P_g(x_d, t) = f(x_d - c_g t)$$

The residual function is therefore

$$r(t) = P_g(x_d, t) - P(x_d, t)$$

if there is one measurment point then, the dimension of the data space is $n = 1$. The source in the adjoint equation is however of the dimension of the state space, which is p dimensional, the conversion is achieved by the conversion matrix K, leading to a vector which has only one non-zero element in it at the element associated with x_d, with the value of $r(t)$.

The adjoint equation is therefore

$$K\mathbf{r}^* + \left(\frac{\partial H}{\partial u}\right)^* u^\dagger = 0$$

with

$$H(\mathbf{u}(m), m) = \left[\frac{\partial^2}{\partial t^2} - m_g^2(x)\nabla^2\right] u(x,t) = 0$$

leading to

$$\left[\frac{\partial^2}{\partial t^2} - m_g^2(x)\nabla^2\right] u^\dagger(x,t) = S(x_d, t)$$

leading to

$$\partial_t \mathbf{s} = \partial_t \begin{bmatrix} u^\dagger \\ v^\dagger \end{bmatrix} = \begin{bmatrix} 0 & I \\ m_g^2\nabla^2 & 0 \end{bmatrix} + \begin{bmatrix} 0 \\ S(x_d, t) \end{bmatrix}$$

$$\partial_t^2 u^\dagger = \partial_t v^\dagger = m_g^2 \nabla^2 u^\dagger + S(x_d, t)$$

solving the equation we get

$$\mathbf{s}(x,t) = e^{Mt} + \int_0^t d\tau e^{M(t-\tau)} \mathbf{S}(\tau)$$

where

$$\mathbf{S}(x_d, t) = \begin{bmatrix} 0 \\ S(x_d, t) \end{bmatrix}$$

$$dt = T_0/(N_t + 1) \rightarrow N_t = \frac{T_0}{dt} - 1$$

Analytical solutions for a 1D wave equation

The equation is

$$u_{tt} - c^2 u_{xx} = S(x,t)$$

with periodic boundary conditions on $[0, L]$
The simplest test case is for a single fourier mode source

$$S(x,t) = A \sin(kx) \cos(\omega t)$$

$k = \frac{2\pi n}{L}$ for $n \in \mathbb{Z}$.
The non-resonant solution ($\omega \neq ck$)

$$u(x,t) = B(t) \sin(kx)$$

where

$$B(t) = \left[C_1 \cos(\omega_k t) + C_2 \sin(\omega_k t) + \frac{A}{\omega_k^2 - \omega^2} \cos(\omega t) \right]$$

with $\omega_k = ck$
For initial conditions $u(x, t=0) = \mathbf{0}$, $u_t(x, t=0) = \mathbf{0}$

$$C_1 = -\frac{A}{\omega_k^2 - \omega^2}$$

$$C_2 = 0$$

$$u(x,t) = \left[\frac{A}{\omega_k^2 - \omega^2} \left(\cos(\omega t) - \cos(\omega_k t) \right) \right] \sin(kx)$$

on resonance $\omega = \omega_k = ck$ we have

$$u_{\text{res}}(x,t) = \left[C_1 \cos(\omega_k t) + C_2 \sin(\omega_k t) + \frac{A}{2\omega_k} t \sin(\omega_k t) \right] \sin(kx)$$

applying the boundary conditions

$$C_1 = 0$$

$$C_2 = 0$$

$$u_{\text{res}}(x,t) = \frac{A}{2\omega_k} t \sin(\omega_k t) \sin(kx)$$

Solution employing the Green function

For such boundary conditions we wrap the solution

$$G_{\text{per}}(x,t) = \frac{1}{2c}\sum_{m=-\infty}^{\infty} H\left(t - \frac{|x+mL|}{c}\right)$$

The solution is a convolution in space and time

$$u(x,t) = \int_0^t \int_0^L G_{\text{per}}(x-\xi, t-\tau)\, S(\xi,\tau)\, d\xi d\tau$$

For a delta function in time (and assuming times where the wave-packet doesn't wrap around)

$$S(x,t) = f(x)\,\delta(t)$$

$$u(x,t) = \frac{1}{2c}\int_{x-ct}^{x+ct} e^{-\frac{(\xi-x_0)^2}{2\sigma^2}} d\xi$$

$$= \frac{1}{4c}\left[\operatorname{erf}\left(\frac{x+ct-x_0}{\sqrt{2}\sigma}\right) - \operatorname{erf}\left(\frac{x-ct-x_0}{\sqrt{2}\sigma}\right)\right]$$

$$u(x,t) = \frac{1}{2c}\int_0^t \int_{x-c(t-\tau)}^{x+c(t-\tau)} S(\xi,\tau)\, d\xi d\tau$$

So for

$$S(x,t) = e^{-\frac{(x-x_0)^2}{2\sigma^2}} e^{-\frac{(t-t_0)^2}{2\tau^2}}$$

$$u(x,t) = \frac{1}{4c}\int_0^t d\tau \left[\operatorname{erf}\left(\frac{x+c(t-\tau)-x_0}{\sqrt{2}\sigma}\right) - \operatorname{erf}\left(\frac{x-c(t-\tau)-x_0}{\sqrt{2}\sigma}\right)\right]$$

Testing the adjoint master equation

If a detector at position x_d records at all times and a constant speed field $m(x) = c$, we have that the predicted signal is $u(x_d, t)$

the residual funciton is then

$$r(x,t) = \left[u_{\text{pred}}(x,t) - u_{\text{obs}}(x,t)\right]\delta_{x,x_d}$$

This enters the adjoint equation as

$$\left(D_\tau^2 - c^2\nabla^2\right) u^\dagger = -K^* r(T-\tau)$$

which can be cast in the form

$$\partial_\tau \left[\begin{array}{c} u \\ v \end{array} \right] = \left[\begin{array}{cc} 0 & I \\ c^2 \nabla^2 & 0 \end{array} \right] \left[\begin{array}{c} u \\ v \end{array} \right] + \left[\begin{array}{c} 0 \\ -r\left(t-\tau\right) \end{array} \right]$$

which has the solution

$$w\left(T\right) = e^{\mathcal{O}\tau} - \int\limits_0^T ds e^{\mathcal{O}s} r\left(T-\tau+s\right)$$